

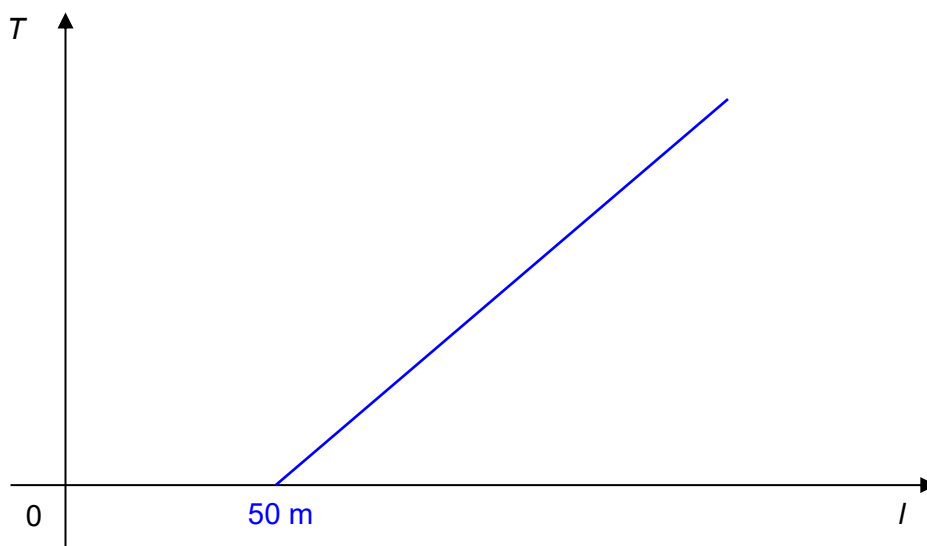
- 2 Fig. 2.1 shows a person of mass 70 kg doing a bungee jump.



Fig. 2.1

The bungee rope has negligible mass and has an unstretched length of 50 m. This bungee rope may be assumed to obey Hooke's law. A force of 200 N extends the bungee rope by 2.5 m. The bungee rope is secured to the feet of the person. The person leans over a bridge above the river and drops vertically downwards towards the river below. You may assume that air resistance is negligible.

- (a) Sketch a graph of the tension T in the bungee rope against its total length l .



[2]

- (b) Show that the elastic potential energy in the bungee rope is directly proportional to the square of its extension e .

Elastic potential energy = Area under Force-extension graph

$$= \frac{1}{2} T e$$

$$= \frac{1}{2} k e (e) \quad \text{since } T = k e \text{ by Hooke's Law}$$

$$= \frac{1}{2} k e^2$$

Hence, elastic potential energy $\propto e^2$ [shown]

[2]

- (c) Determine the elastic potential energy in the bungee rope when the person has reached the lowest point of his jump.

At lowest point, by Conservation of Energy,

Loss in GPE = Gain in Elastic PE

$$mg(l_0 + e) = \frac{1}{2} k e^2$$

$$70(9.81)(50 + e) = \frac{1}{2} (200 / 2.5) e^2$$

$$40 e^2 - 70(9.81) e - 70(9.81)(50) = 0$$

$$e = 39.11 \text{ m}$$

Hence, elastic potential energy = $\frac{1}{2} k e^2$

$$= \frac{1}{2} (200/2.5)(39.11)^2$$

$$= 6.1 \times 10^4 \text{ J}$$

elastic potential energy = J [3]

- (d) Explain why it would be extremely dangerous to have a bungee rope that is much stiffer.

.. If the bungee rope is stiffer, the extension and hence, the time taken Δt for the jumper to change his momentum from a high value to zero (at the lowest point) will be shorter. Therefore, the rate of change of momentum experienced by the jumper will be higher and this will result in a larger force acting on the jumper which may injure the jumper.

[1]

[Total: 8]